Business Intelligence Internship Assignment

Key Insights Derived from the Analysis

Insight 1 — Claim Risk Concentration

Claims are heavily concentrated among vehicles purchased on specific dates (7th, 14th, 21st, and 28th of each month). This simulated defect pattern demonstrates how operational issues can create **clustered claim events**, significantly increasing short-term claim exposure.

Business implication:

Insurance companies must monitor product defect patterns or external events that may trigger sudden claim spikes.

Insight 2 — Longer Policy Tenure Increases Risk Exposure

Policies with longer tenures (3–4 years) show higher cumulative claim probability compared to shorter policies. While longer policies generate more premium revenue, they also expose the insurer to **extended risk duration**.

Business implication:

Insurers must carefully price long-tenure policies to maintain profitability.

Insight 3 — Portfolio Profitability Depends on Loss Ratio

The portfolio's loss ratio (Claims ÷ Premium) indicates the balance between revenue and risk. If claim frequency increases annually (as simulated with a 5% increase), the loss ratio could rise significantly, reducing underwriting profitability.

Business implication:

Continuous monitoring of claim frequency trends is critical for adjusting pricing strategies.

Insight 4 — Future Liability Risk

Vehicles that have not yet filed claims represent **potential future liabilities**. Estimating this exposure helps insurers understand the **true long-term financial risk of their policy portfolio**.